

ORIGINAL ARTICLE

Application of machine learning algorithms for predicting methylglyoxal concentrations in processed packaged foods

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ABSTRACT

Background and aim: Methylglyoxal (MGO) is a glucotoxic reactive α -dicarbonyl metabolite that plays a vital role in processed packaged food products. MGO can be formed both endogenously through glycolysis and exogenously during the thermal processing of foods, particularly via the Maillard reaction. This study aims to propose algorithms for predicting MGO formation in foods; it presents a new concept in the literature, “pMGO”.

Methods: In this study, pMGO analyses were performed via machine learning (ML) algorithms on the basis of food composition data obtained from open-access libraries. Data were collected from peer-reviewed studies published between 2020 and 2025 that reported MGO concentrations measured via validated chromatographic methods (HPLC, UHPLC–MS/MS, and LC–MS), and all values were harmonized to $\mu\text{g}/100\text{ g}$. The predictor variables were glucose, fructose, sucrose, total sugar, carbohydrate, protein, fat, saturated fat, fiber and salt.

Results: CatBoost outperformed the other evaluated models in minimizing prediction error ($\text{RMSE} = 156.75 \pm 64.92\ \mu\text{g}/100\text{ g}$) and was identified as the best-performing algorithm for pMGO estimation. Glucose, fructose and saturated fat were the most effective variables, whereas carbohydrates and protein contributed more indirectly.

Conclusion: Findings suggest that focusing on some components, especially sugar type and saturated fat, can improve the prediction of MGO exposure.

Key words: methylglyoxal, artificial intelligence, prediction of MGOs, catboost, dicarbonyl products



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Introduction

Methylglyoxal (MGO) is a reactive α -dicarbonyl and glucotoxic compound that can be formed in the body not only endogenously but also exogenously (1, 2, 3). It is produced through various reactions, including the interaction of sugars with amino acids during the Maillard reaction, which generates flavor compounds but also leads to the formation of toxic byproducts. This compound, which emerges particularly during the Maillard reaction (MR), is considered one of the most important precursors of advanced glycation end products (AGEs) (1).

MGO is commonly found in processed, baked, and sugary foods and beverages; it is particularly common in sweets, baked goods, some beverages, and heat-treated meats and fish (1). Production/processing conditions, the type of sugar used, storage, reheating, and digestion processes can significantly alter MGO levels (1, 2). Proteins, amino acids, and plant phenolic compounds can both contribute to MGO formation and reduce it by scavenging it, offering potential for reduction strategies (3, 4). Overall, studies are beginning to reveal where MGO is found, how it is formed and transformed in foods and during digestion, and how to assess the effects of dietary MGO on health (4).

From a health perspective, the accumulation of MGOs in metabolism enhances the risk of mutations and triggers inflammation (4). Moreover, MGO is also associated with the onset and progression of several pathologies, such as diabetes, cancer, and liver and kidney diseases. Furthermore, reducing MGO levels may represent a potential strategy for disease prevention, and it may also be evaluated as a biomarker to estimate disease risk and progression (5). The detoxification of MGO in the body involves the glyoxalase system, aldehyde dehydrogenases, and aldose reductases; dysfunctions in these systems may contribute to diabetic complications (5, 6). Additionally, bioactive components such as resveratrol, anthocyanins, and polyphenols are effective detoxification mechanisms for protecting against the negative effects of MGO (7).

The main analytical methods used for the determination of MGO include high-performance liquid chromatography (HPLC), gas chromatography–mass spectrometry (GC–MS), ultrahigh-performance liquid

chromatography–tandem mass spectrometry (UH-PLC–MS/MS) (7, 8), liquid chromatography–mass spectrometry (LC–MS) (9), ultraviolet–visible spectroscopy (UV–Vis) spectrophotometry, and immunochemical methods (ELISAs) (10, 11). Specifically, ELISA operates on the basis of a signal generation principle that depends on MGO inhibiting a substrate–enzyme reaction. Nevertheless, these conventional methods have certain limitations in terms of accuracy and sensitivity, speed and efficiency, ease of application, field suitability, and cost when evaluated under the framework of emerging diagnostic technologies and quality control processes.

Recently, the need for data-driven strategies has become increasingly evident. In particular, in prediction, classification, and decision support systems, ML has become a game-changing tool (12). For example, while traditional statistical methods have been inadequate for evaluating soybean varieties and predicting soymilk quality, a stochastic gradient descent-based deep learning model has developed and successfully predicted both yield and quality on the basis of physicochemical properties. In addition, another study demonstrated the potential of ML models in predicting AGE formation and its health impacts (12, 13). Moreover, Molecular Transformer–ML algorithms have provided significant insights into possible reactions under different thermal conditions, and their applications in fields such as bioinformatics and microbiology are becoming increasingly clear (14, 15). In this context, the accurate and rapid prediction of MGO, which is important in terms of both its metabolic effects in biological systems and its presence in processed foods, has become critical for food safety and public health. This study proposes a predictive tool called “predicted MGO” (pMGO), which estimates MGO levels via components such as glucose, fructose, sucrose, total sugars, carbohydrates, protein, fat, saturated fat, fiber, and salt through models such as Ridge, Lasso, ElasticNet, random forest, gradient boosting, XGBoost, CatBoost, AdaBoost, and artificial neural network (ANN). This study aims to estimate the risk of foodborne MGOs on the basis of component-level effects (nutritional values) while reducing the cost and laboratory workload associated with the traditional MGO analyses mentioned above.

Methods

Data collection and bibliometric study

In this study, all the data included in the analyses were compiled from peer-reviewed studies indexed in the Web of Science Core Collection (WoS), Scopus, Google Scholar, and ResearchGate, following a structured literature screening that was completed on October 20, 2025. Keyword-based searches were performed via the term “methylglyoxal” in combination with food-related and analytical descriptors. Studies were included if they (i) were published in English between 2020 and 2025, (ii) reported quantitative MGO concentrations in processed packaged foods, (iii) used validated chromatographic methods (HPLC, UHPLC-MS/MS, or LC-MS), and (iv) provided either the raw nutritional composition of the analyzed foods or sufficient information to recover it from the product label. Studies were excluded if they reported only relative values, if MGO concentrations were below the limit of detection without numerical reporting, or if the food matrix could not be matched to a clear processed food category. When the same food product

appeared in more than one source, the most recent and methodologically detailed study was retained. All MGO values were harmonized to a common unit ($\mu\text{g}/100\text{ g}$) before integration. The overall workflow of data acquisition, preprocessing, model training, and prediction is summarized in Figure 1, which depicts the inputs (large-scale datasets, processed food categories, and MGO-related literature), the data-processing pipeline (cleaning, normalization, feature engineering), the machine-learning algorithms evaluated, and the intended applications of the resulting pMGO tool.

Sample preparation and analysis

Jupyter notebooks, a web application software for machine learning with Python 3.11.5, were used to process the collected data. The following libraries were integrated: Pandas, NumPy, Matplotlib, Scikit-Learn, xgboost, lightgbm, and catboost. The analysis was performed on a desktop PC that was configured with 32.00 GB of RAM, a 3.70 GHz Intel Core i7-8700K processor, and an NVIDIA Titan XP graphics card, all of which were operating Windows 10.

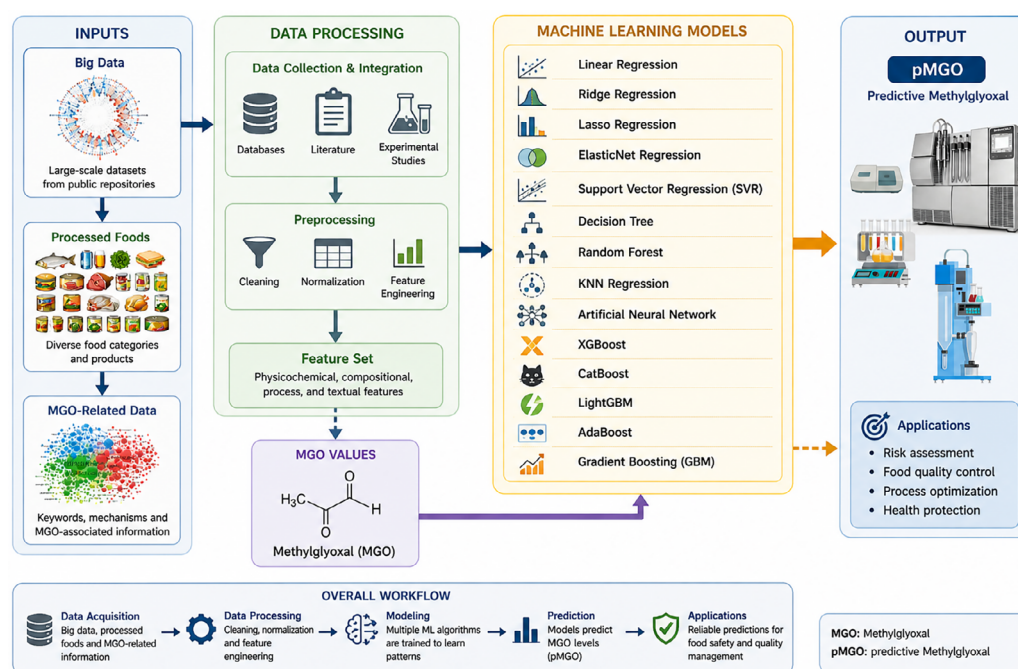


Figure 1. Methodological flow chart of the study.

Variable description

Carbohydrate-rich foods, particularly those with high fructose and glucose contents, promote the formation of MGO through the Maillard reaction. Fructose, in particular, has been observed to trigger MGO production at higher levels than glucose does. Proteins react with carbonyl compounds such as MGO, contributing to the formation of AGEs; thus, reactive carbonyl species tend to appear at elevated levels in protein-rich foods. Similarly, in products with a high saturated fat content, MGO levels increase due to lipid peroxidation. High-fiber foods are digested more slowly, reducing the glycemic response, and have a limiting effect on MGO formation; this effect is particularly notable in foods such as oats. The amount of salt has a limited direct effect on MGO formation; however, high salt intake may indirectly contribute to AGE production by promoting metabolic pathways. Simple sugars such as fructose, glucose, and sucrose contribute to MGO production both directly and indirectly; in particular, sucrose catalyzes this process, breaking down it into fructose and glucose. As total sugar values and thermally processed/ultra-processed foods increase, MGO levels rise significantly, and fructose plays a pivotal role in this increase. Considering all these changes, selected studies from the literature have considered auxiliary nutritional components in the MGO estimation process.

Machine learning methods

Machine learning (ML) techniques, which are gaining increasing importance in food science, have been used. Because ML algorithms can model nonlinear relationships and improve prediction accuracy, several supervised learning methods have been used to estimate MGO content. In this context, algorithms such as linear regression (LR), ridge regression (RR), LASSO regression, ElasticNet regression, support vector regression (SVR), decision tree, random forest, K-nearest neighbors (KNN), artificial neural networks (ANNs), XGBoost, CatBoost, LightGBM, AdaBoost, and gradient boosting machines (GBMs) were evaluated.

Each model was compared in terms of its predictive power, interpretability, computational efficiency, and ability to address domain-specific challenges,

such as high dimensionality, multicollinearity, and nonlinear interactions. Through this methodological approach, we aimed to identify the most suitable algorithms for accurately predicting MGO levels in food products. Table 1 summarizes the ML algorithms employed in this study. Each algorithm addresses different aspects of the prediction tasks. Linear models (LR, RR, Lasso, ENR) were included as interpretable baselines to evaluate the impact of L1/L2 regularization on multicollinearity arising from correlated nutritional composition variables. Tree-based and ensemble methods (DT, RF, XGBoost, CatBoost, LightGBM, AdaBoost, and GBM) were applied to capture the nonlinear interactions among the carbohydrate, protein, and fat contents that are mechanistically linked to MGO formation during thermal processing. Kernel-based (SVR), instance-based (KNN), and neural network (ANN) approaches were further incorporated to benchmark fundamentally different learning paradigms against the ensemble methods used. This combination of models allowed direct comparison and helped identify which algorithm performed best for estimating MGO concentrations in processed packaged foods.

Performance evaluation metrics

The effectiveness of the model was systematically assessed via a range of statistical measures, especially the coefficient of determination, along with the mean absolute error (MAE) and the root mean square error (RMSE), which were applied to both the validation and testing datasets. The coefficient of determination (R^2) serves as a vital metric that measures the extent to which the variability in the dependent variable is explained by the predictive model. On the other hand, the RMSE offers a significant evaluation of the standard deviation of the residuals, thus providing a numerical measure of the model's prediction accuracy. The mean absolute error (MAE) offers a valuable representation of the average size of the differences between the predicted and actual values, facilitating a clearer understanding of the model's predictive performance. In a study where relative error metrics were applied to evaluate model accuracy, a relative error of less than 20% and a relative error standard deviation of less than

Table 1. Methods for ML applications in this study

Model	Description	Cited
LR	Models the relationship between independent variables and a continuous outcome by fitting a linear equation that minimizes the sum of squared errors; valued for simplicity and interpretability.	(12)
RR	Applies L2 regularization to reduce multicollinearity, improving model stability and generalization in high-dimensional datasets.	(16)
LR	Uses L1 regularization to shrink some coefficients to zero, enabling effective variable selection and improving interpretability.	(17)
ENR	Combines L1 and L2 regularization to address multicollinearity while performing variable selection in complex datasets.	(18)
SVR	A kernel-based method that models nonlinear relationships by maximizing margin within a defined error tolerance.	(15)
DT	A rule-based model (CART) capable of capturing nonlinear patterns with high interpretability and minimal preprocessing.	(19)
RF	An ensemble method that aggregates multiple decision trees to improve accuracy and reduce overfitting.	(15)
KNN	A nonparametric approach that predicts values based on the average of nearest neighbors in feature space.	(20)
ANN	A layered computational model inspired by the human brain, capable of capturing complex nonlinear relationships.	(21)
XGBoost	An efficient gradient boosting algorithm that builds trees sequentially using optimization techniques, offering high accuracy and robustness.	(22)
CatBoost	A gradient boosting algorithm optimized for handling categorical variables and reducing overfitting.	(23)
LightGBM	A high-performance boosting framework using histogram-based learning and optimization techniques (GOSS, EFB) for speed and efficiency.	(24)
AdaBoost	An adaptive boosting algorithm that iteratively combines weak learners by focusing on hard-to-predict instances.	(25)
GBM	Builds models sequentially by minimizing residual errors, providing high accuracy and flexibility for prediction tasks.	(26)

Artificial Neural Network: ANN; ElasticNet Regression (ENR); Gradient Boosting (GBM); Decision Tree (DT); KNN Regression (KNN); Lasso Regression (LR); LightGBM; Linear Regression (LR); Random Forest (RF); Ridge Regression (RR); Support Vector Regression (SVR); XGBoost

40% were obtained. This finding indicates that the predictive capacity of the model was within acceptable error margins (12). In a related study, the model achieved a 76% top-3 accuracy across the full dataset, with four out of five selected reactions being consistent with the experimental validation. This finding underscores the high accuracy of the model in practical scenarios. When evaluating regression models, metrics such as the coefficient of determination (R^2), cross-validated R^2 (Q^2), mean absolute error (MAE), root mean square error (RMSE), and fold ratio are commonly utilized(14). These metrics collectively offer a

comprehensive evaluation of the precision and reliability of the model. In the assessment of regression models, the fold ratio was employed to analyze the ratios within 2-fold and 3-fold error margins, providing insights into the model's predictive capabilities (15). In a separate investigation centered on predicting the glycemic index, the CatBoost algorithm outperformed other models by securing the highest test R^2 value (0.977), along with the lowest RMSE (1.674) and MAE (1.302) (27). Another study assessed the predictive power of different statistical methods by utilizing the R^2 coefficient and standard deviation

between observed and predicted phenotypes. In various cross-validation scenarios, the standard deviation was derived from the outcomes of ten repetitions, and these metrics were used to evaluate the reliability of the model (28). In this analysis, models that demonstrated high R^2 values, indicating strong explanatory capabilities with low RMSE values, were deemed to perform optimally. This systematic evaluation process highlights the importance of these statistical metrics in assessing the effectiveness of the model.

Hyperparameter tuning for model optimization

In this study, the algorithm exhibiting the best performance among the basic models was first determined, and hyperparameter tuning was applied for

further optimization. To fine-tune the chosen model for optimal performance across various data representations, this tuning procedure was applied to the three best-performing datasets as well as their feature-selected and feature-extracted counterparts.

Predictive models were created in this study via a total of 14 machine learning algorithms. The grid search method was used for hyperparameter tuning to maximize each model's performance. A predetermined hyperparameter search space was built for each model via library defaults and best practices from the literature. Table 2 displays the entire set of evaluated hyperparameter ranges. Throughout the search process, 10-fold cross-validation was employed to guarantee robustness and prevent overfitting. By averaging performance metrics across several validation folds, this approach made it possible to estimate model

Table 2. Model parameters for hyperparameters

LR	fit_intercept: [True, False]			
RR	alpha: [0.01, 0.1, 1.0, 10.0]	solver: ['auto', 'svd', 'cholesky']		
LR	alpha: [0.01, 0.1, 1.0, 10.0]	max_iter: [1000, 2000]		
ElasticNet Regression	alpha: [0.01, 0.1, 1.0]	l1_ratio: [0.1, 0.5, 0.9]	max_iter: [1000, 2000]	
SVR	C: [0.1, 1, 10]	kernel: ['linear', 'rbf']	gamma: ['scale', 'auto', 0.1]	
Decision Tree	max_depth: [3, 5, 10, None]	min_samples_split: [2, 5, 10]	min_samples_leaf: [1, 2, 4]	
Random Forest	n_estimators: [50, 100, 200]	max_depth: [3, 5, 10, None]	min_samples_split: [2, 5]	
KNN	n_neighbors: [3, 5, 7, 9]	weights: ['uniform', 'distance']	p: [1, 2]	
ANN	hidden_layer_sizes: [(50), (100), (50, 50)]	alpha: [0.0001, 0.001, 0.01]	learning_rate_init: [0.001, 0.01]	learning_rate: ['constant', 'adaptive']
XGBoost	max_depth: [3, 5, 7]	learning_rate: [0.01, 0.05, 0.1]	n_estimators: [50, 100, 200]	subsample: [0.8, 1.0]
CatBoost	depth: [4, 6, 8]	learning_rate: [0.01, 0.05, 0.1]	iterations: [50, 100, 200]	l2_leaf_reg: [1, 3]
LightGBM	max_depth: [3, 5, 7]	learning_rate: [0.01, 0.05, 0.1]	n_estimators: [50, 100, 200]	num_leaves: [20, 31, 40]
AdaBoost	n_estimators: [50, 100, 200]	learning_rate: [0.01, 0.1, 1.0]	loss: ['linear', 'square']	
Gradient Boosting	n_estimators: [50, 100, 200]	max_depth: [3, 5, 7]	learning_rate: [0.01, 0.05, 0.1]	subsample: [0.8, 1.0]

generalizability with accuracy. The tuning strategy used was grid search, which methodically investigated a predetermined range of hyperparameter combinations.

Two main criteria were used to choose the best model for each algorithm: (i) the model’s explanatory power, or highest coefficient of determination (R^2), and (ii) the model’s prediction accuracy, or lowest root mean squared error (RMSE). The best-performing hyperparameter combination for each algorithm was determined by computing these metrics across all cross-validation folds. This tuning and evaluation process improved the reliability and comparability of the results.

Results

Statistical insight

Strong positive correlations were observed between glucose and fructose, between fructose and total

sugars, and between glucose and total sugars (with correlation coefficients close to 1; Figure 2, Table 3 and Table 4). This finding is highly meaningful, as all three variables belong to the same carbohydrate family, and total sugar is composed of these constituents. The significant positive relationship between the sucrose content and total sugar content demonstrated the major role of sucrose in determining the overall sugar content.

The strong positive correlation between total fat and saturated fat indicates that saturated fat makes up a large portion of total fat. On the other hand, a moderate positive correlation was observed between MGO and both saturated fat and fructose, suggesting that MGO levels might fluctuate on the basis of the presence of certain sugars or saturated fats. A weak-to-moderate positive relationship was found between MGO and total sugars, implying that an increase in sugar content could result in higher MGO levels.

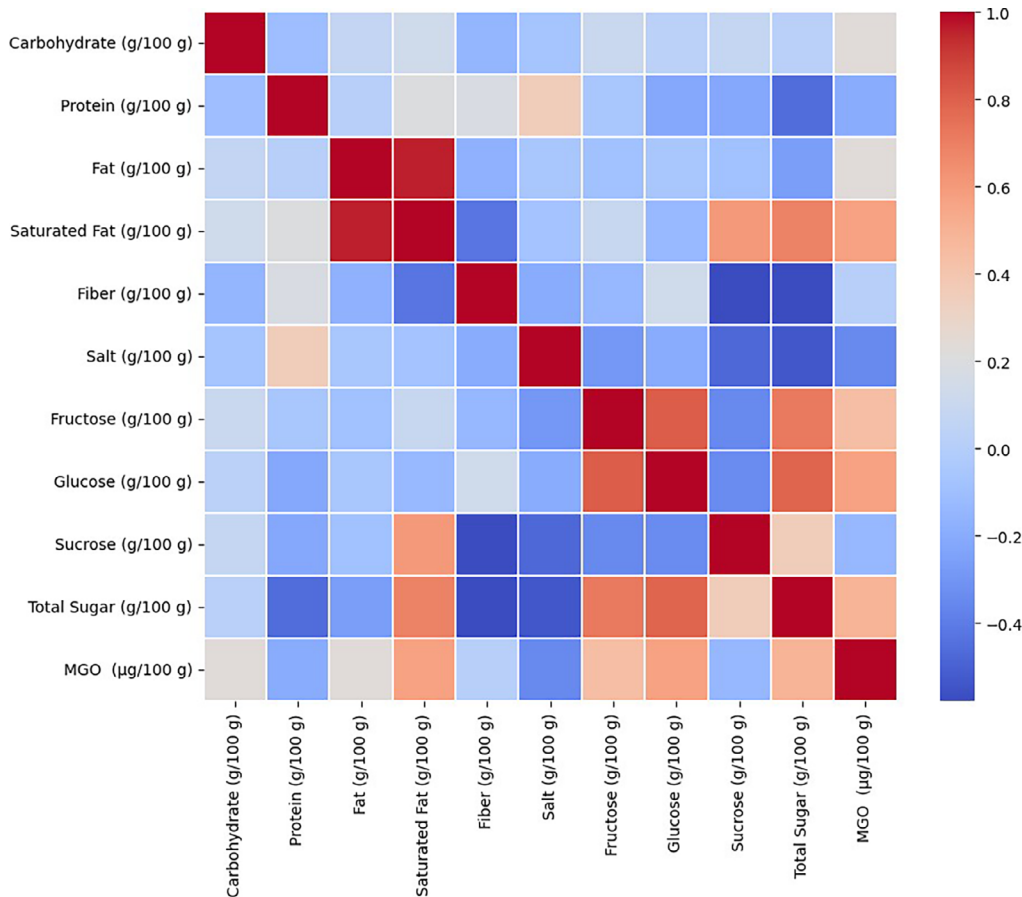


Figure 2. Heatmap of MGO.

Table 3. Summary statistics for nutritional values and MGOs

	Carbohydrate (g/100 g)	Protein (g/100 g)	Fat (g/100 g)	Saturated Fat (g/100 g)	Fiber (g/100 g)	Salt (g/100 g)	Fructose (g/100 g)	Glucose (g/100 g)	Sucrose (g/100 g)	Total Sugar (g/100 g)	MGO (μ g/100 g)
count	192	227	242	54	51	54	226	226	236	178	453
mean	60.4	7.3	12.5	4.6	5.1	1.6	6.5	15.1	15.7	38.1	569.6
std	54.5	6.2	11.7	3.7	2.9	1.5	6.7	16.6	14.8	23.5	1547.5
min	4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
25%	48.8	3.0	2.8	1.7	3.0	0.7	0.2	0.5	5.4	17.6	42.0
50%	61.3	5.9	10.4	3.2	4.9	1.2	3.2	5.2	12.3	34.7	97.0
75%	71.8	9.7	20.0	7.2	6.4	1.8	12.3	31.0	21.4	62.5	358.0
max	757.0	32.9	98.0	16.8	13.2	7.5	26.0	48.9	65.8	74.5	14132.7

Table 4. Correlation matrix of nutritional values

	Carbohydrate (g/100 g)	Protein (g/100 g)	Fat (g/100 g)	Saturated Fat (g/100 g)	Fiber (g/100 g)	Salt (g/100 g)	Fructose (g/100 g)	Glucose (g/100 g)	Sucrose (g/100 g)	Total Sugar (g/100 g)	MGO (μ g/100 g)
Carbohydrate (g/100 g)	1	-0.11	0.07	0.13	-0.15	-0.07	0.1	0.03	0.08	0.02	0.23
Protein (g/100 g)	-0.11	1	0.01	0.2	0.18	0.35	-0.06	-0.22	-0.22	-0.46	-0.2
Fat (g/100 g)	0.07	0.01	1	0.96	-0.17	-0.06	-0.09	-0.06	-0.09	-0.26	0.23
Saturated Fat (g/100 g)	0.13	0.2	0.96	1	-0.43	-0.08	0.09	-0.13	0.6	0.69	0.57
Fiber (g/100 g)	-0.15	0.18	-0.17	-0.43	1	-0.2	-0.14	0.13	-0.58	-0.58	0.01
Salt (g/100 g)	-0.07	0.35	-0.06	-0.08	-0.2	1	-0.29	-0.2	-0.48	-0.54	-0.35
Fructose (g/100 g)	0.1	-0.06	-0.09	0.09	-0.14	-0.29	1	0.81	-0.35	0.72	0.44
Glucose (g/100 g)	0.03	-0.22	-0.06	-0.13	0.13	-0.2	0.81	1	-0.34	0.79	0.57
Sucrose (g/100 g)	0.08	-0.22	-0.09	0.6	-0.58	-0.48	-0.35	-0.34	1	0.35	-0.14
Total Sugar (g/100 g)	0.02	-0.46	-0.26	0.69	-0.58	-0.54	0.72	0.79	0.35	1	0.49
MGO (μ g/100 g)	0.23	-0.2	0.23	0.57	0.01	-0.35	0.44	0.57	-0.14	0.49	1

Negative correlations were noted between dietary fiber, glucose, and sucrose, suggesting that foods rich in fiber might have lower levels of simple sugars. Similarly, the negative correlation between salt and sugar components is significant, supporting the notion that salty foods generally contain less sugar. These findings offer valuable insights into the structural and functional relationships between nutrients and are crucial for conducting thorough nutritional analyses.

Model application and evaluation of scenarios

Table 5 suggests that carbohydrates play a significant role in MGO formation. The average carbohydrate content was 60.4 g per 100 g, while the mean MGO concentration was 185.3 μ g per 100 g. Both variables exhibited similar distribution patterns, with a carbohydrate standard deviation of 54.5 and an MGO standard deviation of 266.1. The maximum values were

Table 5. Carbohydrate, protein, and fat effects on MGO formation (S1)

	Carbohydrate (g/100 g)	Protein (g/100 g)	Fat (g/100 g)	MGO (µg/100 g)
count	192	192	192	192
mean	60.4	8.3	13.9	185.3
std	54.5	6.3	11.9	266.1
min	4.0	0.5	0.0	1.6
0.25	48.8	4.1	3.0	33.1
0.5	61.3	6.6	11.8	74.2
0.75	71.8	10.5	21.2	205.8
max	757.0	32.9	98.0	1573.0

757 g for carbohydrates and 1573 µg for MGO. These data imply that reducing sugars, especially fructose and glucose, are directly responsible for initiating MGO formation via the Maillard reaction. The average protein content was 8.3 g/100 g, which was significantly lower than the carbohydrate content. While proteins do not directly generate MGOs, they interact with existing MGOs and contribute to AGE development. The broad range of MGO levels contrasts with the more consistent protein levels, indicating that proteins indirectly affect MGO levels. The average fat content is 13.9 g/100 g and may play a role in MGO formation through lipid peroxidation. However, this effect is much more limited than that of carbohydrates. The influence of fat on MGO may only be significant in certain fat-rich products. In conclusion, carbohydrates are the most dominant and decisive component in MGO formation, whereas proteins and fats are considered indirect or supportive factors.

According to Table 6, the average MGO (methylglyoxal) value is 243.7 µg/100 g, reaching a notably high maximum of 1022.6 µg. The standard deviation of 252.2 indicates considerable variability in the data, suggesting that certain products play a much greater role in MGO production. The average carbohydrate content was 73.3 g, which increased in parallel with MGO, confirming that carbohydrates are a key determining component. When sugar types are examined, although fructose is present at low levels (average 0.5 g), it is known to trigger MGO formation more strongly than glucose does, indicating its critical role

in MGO formation. Glucose had an average value of 1.0 g, whereas sucrose had the highest average value of 23.2 g. Sucrose is a disaccharide that can be broken down into glucose and fructose, serving as an indirect yet significant source of MGO formation. Protein (5.8 g) and fat (19.7 g) levels contribute to MGO formation in a supportive role, but they are not as significant as carbohydrates, particularly sugars. Proteins primarily contribute to the formation of AGEs by reacting with MGO, while fats may promote MGO formation through lipid peroxidation.

This is especially true under high-heat conditions. Therefore, in MGO prediction processes, the type and amount of sugar are highly important for model performance.

In the Table 7 dataset, carbohydrates emerged as the most significant nutrient influencing MGO formation. Carbohydrates, averaging 96.8 g, are the most dominant component and include reducing sugars such as fructose, glucose, and sucrose, which closely correlate with the MGO distribution. When individual sugar types were examined, the average amounts were as follows: total sugar at 13.8 g, sucrose at 12.3 g, fructose at 0.6 g, and glucose at 0.9 g. These figures suggest that sucrose is the predominant sugar, indirectly contributing to MGO production by breaking down into fructose and glucose. Despite being present in small amounts, fructose has greater potential to generate MGO than glucose does, making it a critical component. Protein (7.5 g) and fat (14.1 g) are present at moderate levels and play indirect roles in

Table 6. Effects of carbohydrates, protein, fat, glucose, fructose and sucrose on MGO formation

	Carbohydrate	Protein	Fat	Fructose	Glucose	Sucrose	MGO
count	87	87	87	87	87	87	87
mean	73.3	5.8	19.7	0.5	1.0	23.2	243.7
std	75.3	3.0	12.2	0.9	1.8	18.6	252.2
min	19.3	0.5	3.0	0.0	0.0	0.0	11.0
25%	58.2	3.1	11.6	0.0	0.0	12.8	64.3
50%	65.0	5.7	16.9	0.1	0.4	18.8	114.6
75%	73.5	8.0	27.0	0.5	1.1	31.0	384.2
max	757.0	16.0	98.0	4.8	8.9	65.8	1022.6

All nutrition values were given as g/100 g and MGO ($\mu\text{g}/100\text{ g}$)

Table 7. All variables affect MGO formation

	Carbohydrate	Protein	Fat	Saturated Fat	Fiber	Salt	Fructose	Glucose	Sucrose	Sugar	MGO
count	24	24	24	24	24	24	24	24	24	24	24
mean	96.8	7.5	14.1	7.2	4.9	1.3	0.6	0.9	12.3	13.8	320.6
std	140.7	2.6	4.6	3.2	2.6	1.6	0.5	1.4	8.5	8.4	286.1
min	56.0	4.0	6.3	2.4	1.5	0.1	0.0	0.0	0.0	0.6	39.9
25%	63.9	5.7	10.6	4.6	2.8	0.7	0.3	0.2	2.0	3.9	72.3
50%	69.2	7.1	14.1	7.0	5.0	0.9	0.5	0.3	16.5	18.0	219.3
75%	73.5	9.0	17.4	9.2	5.8	1.2	0.7	0.9	18.7	19.7	450.3
max	757.0	16.0	22.2	16.8	11.5	7.5	1.9	6.4	24.9	25.0	1010.6

All nutrition values were given as g/100 g and MGO ($\mu\text{g}/100\text{ g}$)

MGO formation. Proteins contribute to the formation of AGEs, whereas fats and saturated fats can increase MGO levels through lipid peroxidation under high-temperature conditions. Fiber (4.9 g) may prevent MGO formation by slowing digestion and reducing the glycemic response. Although salt (1.3 g) does not have a direct effect, it may indirectly contribute to MGO formation through its influence on metabolic processes. In conclusion, carbohydrates, particularly sugar types, are the primary factors influencing MGO levels in the S3 dataset, whereas other nutrients mostly play supportive or limiting roles.

Tables 5, 6, and 7 show that carbohydrate and sugar contents are important factors in MGO formation.

Tables 5 and 6, in particular, demonstrate the effect of reducing sugars such as glucose, fructose, and sucrose, suggesting a significant role for carbohydrates. The study results show that despite its small amount, fructose is highly reactive. In contrast, sucrose contributes indirectly by breaking down into glucose and fructose. Table 7 presents a multivariate structure in which all the variables (carbohydrates, sugar types, protein, fat, fiber, and salt) are analyzed together. These findings indicate that MGO levels can best be explained by the type and amount of sugar. In summary, all three tables show that sugar types and carbohydrate levels are primary determinants of MGO formation, whereas other components play supporting or indirect roles.

Machine learning modeling and model performance

Table 8 compares the regression performance of various machine learning algorithms across three datasets (Table 5, Table 6, and Table 7). The assessment metrics included the coefficient of determination (R^2) and root mean square error (RMSE). The R^2 value indicates the model's explanatory power with respect to the dataset, whereas the RMSE represents the average magnitude of the prediction error. Together, these metrics provide a comprehensive assessment of the model's precision and ability to generalize. The results show that each model's performance varies depending on the characteristics of the dataset used. The findings indicate that the CatBoost algorithm produced the best outcomes for the S2 dataset. Compared with the other models, CatBoost outperformed the other models, achieving an R^2 value of 0.52 ± 0.30 and an RMSE of 156.75 ± 64.92 for accuracy and error minimization. This performance is consistent with CatBoost's known capacity for handling categorical variables and reducing overfitting. The model's success was further enhanced by precise hyperparameter tuning and accurate validation via 10-fold cross-validation. The Lasso and ElasticNet regression models exhibited superior performance on the S3 dataset, with R^2 values of approximately 0.54. However, their RMSE values remained elevated compared with those of CatBoost

on S2, suggesting a potential inadequacy in reducing prediction errors despite greater explanatory power. Ensemble methods, including random forest, AdaBoost, and gradient boosting, typically exhibit reduced R^2 values and increased RMSEs. These findings indicate that these models may be at risk of overfitting or lack adequate adaptability to the specific characteristics of the datasets used. Overall, model performance depended not only on the algorithm but also on the dataset, the hyperparameter ranges, and the evaluation strategy. When selecting a model, it is vital to consider the average performance metrics, variance, and generalizability. In practical scenarios, a model that records the highest R^2 or lowest RMSE on a given dataset may not necessarily be the best fit for other datasets. The modeling process necessitates a multidimensional assessment and a carefully structured validation approach.

Alignment of variable importance with findings in the scientific literature

According to the ridge regression model, fructose was the most influential factor in promoting MGO formation, with saturated fat and glucose also playing significant roles, as highlighted in this study. In contrast, substances such as fat, fiber, and salt were observed to mitigate MGO levels. The findings suggested that some specific nutrients, such as sugar type

Table 8. Performance results of the best-performing models

	Model	Dataset	R^2 Mean $\pm$ Std	RMSE Mean
1	Ridge Regression	S3	0.47 ± 0.23	190.23 ± 50.66
2	Lasso Regression	S3	0.54 ± 0.29	172.98 ± 63.13
3	ElasticNet Regression	S3	0.54 ± 0.25	177.05 ± 55.93
4	Random Forest	S2	0.43 ± 0.31	175.93 ± 75.42
5	KNN Regression	S2	0.40 ± 0.45	170.29 ± 80.83
6	KNN Regression	S3	0.48 ± 0.25	187.78 ± 56.48
7	XGBoost	S2	0.42 ± 0.33	176.01 ± 78.64
8	CatBoost	S2	0.52 ± 0.30	156.75 ± 64.92
9	CatBoost	S3	0.47 ± 0.29	189.05 ± 60.54
10	AdaBoost	S2	0.47 ± 0.31	167.20 ± 68.99
11	AdaBoost	S3	0.46 ± 0.33	188.68 ± 68.87
12	Gradient Boosting	S2	0.42 ± 0.37	178.21 ± 85.19

and saturated fat, in the model may enhance its predictive accuracy. These results are also supported by data-driven modeling studies in the literature. For example, ridge regression-based linear models have been used to predict protein N-glycosylation, reducing multicollinearity and achieving successful results with a 74% Kolmogorov–Smirnov statistic (29). In addition, in another study, four various machine learning algorithms were compared to predict glycation concentrations on the basis of fluorescence properties. Studies have demonstrated that ridge regression and other data-driven methods can provide high accuracy in predicting processes, in line with the findings obtained in this study for MGO prediction (16). The results of the ridge regression in this study are consistent with those of previous studies that identified the model as a reliable and interpretable predictive tool in food science. A previous study successfully predicted the bulgur yield from wheat properties; linear regression was used to analyze microbiological and chemical data (30, 31). High accuracy is achieved with interval ridge regression in spectral data, demonstrating its effectiveness in complex systems. Findings have shown that ridge regression can deliver strong predictive performance with appropriate variable selection (32).

The LASSO regression model used in this study offered a particular approach, producing results in variable selection and interpretation related to MGO formation. This model identified fructose, glucose, and saturated fat as the most significant variables in MGO formation. While total sugar supported the process, fiber and salt had limiting effects, and sucrose, carbohydrate, fat, and protein were excluded from the model. This clearly illustrates both the dominant role of fructose. Likewise, the literature shows that LASSO regression is effective in analyzing biological and chemical data. One study successfully applied Lasso to predict vitamin C and reducing sugars in sweet potatoes via hyperspectral imaging (17). The other lasso-based methods model chemical changes during Baijiu liquor aging, demonstrating high accuracy with a Lasso-based method (33). These studies confirm that Lasso has strong predictive power in complex systems and aligns with the findings of this study. The ElasticNet regression model used in this study identified fructose, saturated fat, and glucose as the most

dominant variables affecting MGO formation. While total sugar supported this effect, components such as fiber, salt, and fat negatively influenced MGO levels. It is thought that fiber may suppress MGO formation by lowering the glycemic response (7). Some variables, such as carbohydrates, proteins, and sucrose, made weak or indirect contributions to the model. These results indicate that fructose and saturated fat should be considered primary determinants in MGO prediction. ElasticNet's strong capacity for variable selection and interpretation is also supported by similar applications in the literature. A study combined ElasticNet with a genetic algorithm for more efficient feature selection in high-dimensional datasets. Their two-layered method reduced model complexity by eliminating irrelevant variables and improved prediction accuracy (34). Similarly, another study developed an elastic net model to predict pesticide residues in tomato products, using physicochemical variables to achieve high accuracy and contribute to model simplification (18). In this context, ElasticNet regression has proven to be an effective method in this study, both in distinguishing the effects of food components on MGO and in managing multiple and correlated variables. It also offers a flexible modeling approach that is consistent with other applications in agriculture and food research.

Recent studies have shown that the random forest is a powerful tool for predicting food components, offering high accuracy, an eco-friendly approach, and computational efficiency. These studies emphasize the advantages of random forests in the development of fast and cost-effective alternatives to traditional chemical analyses (35, 36). Accordingly, the findings of this study align with those in the literature, where random forest was used to predict MGO formation. Carbohydrates, especially those containing reducing sugars, were the most decisive variable, contributing approximately 40% to MGO prediction. Components, including glucose and protein, emerged as important factors, providing direct and indirect contributions, respectively. Sucrose and fructose had moderate contributions, whereas the effect of fat was limited. This finding indicates that the random forest algorithm is both effective in overall model accuracy and in identifying variable importance. The findings of this study confirm, similar to the results of (35) and (36), that

random forest is an effective predictive tool in different food matrices and has broad application potential in food science. This model can perform successfully in complex biochemical systems, not only in terms of accuracy but also in terms of interpretability.

According to XGBoost, glucose was the primary factor influencing MGO formation, with carbohydrates and fructose also playing significant roles. Sucrose and protein contributed moderately, whereas fat had a minimal effect. The model particularly highlighted the predictive ability of various sugar types and demonstrated exceptional precision in differentiating between nutritional components within the validation set. These findings are consistent with other documented uses of XGBoost in the literature. For example, Feng et al. successfully established the components affecting sediment formation in ginseng beverages, whereas Wiens et al. demonstrated that XGBoost achieved high accuracy in analyzing large datasets and biological systems. Both studies underscore the model's reliability in making predictions and selecting variables in complex systems (22, 37). In this study, the CatBoost model proved to be an effective tool for determining the significance of variables affecting MGO formation. The model highlighted glucose, saturated fat, and total sugar in various forms as the most important predictors. Given the direct influence of glucose, the impact of saturated fat under heat, and the overall sugar load, these variables significantly enhanced the prediction accuracy. Conversely, fructose, sucrose, and salt were moderately effective, whereas carbohydrates, fiber, and fat had only minor impacts. These findings underscore that the type of sugar is more influential than the total carbohydrate content in the formation of MGO.

CatBoost has shown superior performance compared with models such as XGBoost and LightGBM in detecting allergen cross-contamination in the food industry with 75% accuracy, thus aligning with findings from other studies (23). CatBoost's ability to handle categorical data and calculate variable contributions makes it useful for both allergen risk analysis and biochemical modeling, as shown in this study.

Variable importance analysis via AdaBoost revealed that glucose was the most influential variable in MGO formation, followed by saturated fat and

fructose levels. These findings suggest that specific sugar types (glucose and fructose) and saturated fat are more decisive in MGO prediction than the total carbohydrate or sugar content. The selective nature of AdaBoost highlights its ability to deliver high discriminative power by focusing on specific components. The AdaBoost algorithm has demonstrated high accuracy and practicality in rapid and sensitive detection tasks (38). When gradient boosting was applied, carbohydrates emerged as the most significant factor influencing MGO formation, followed by glucose and protein. Carbohydrates account for the total sugar content, and glucose is crucial for MGO production owing to its direct effects. Sucrose and fructose contributed moderately, whereas fat had the least influence on the models. These results demonstrate that the gradient boosting algorithm prioritizes sugar types and total carbohydrate content in MGO prediction. Similarly, in another study, gradient boosting-based methods (particularly CatBoost and LightGBM) successfully selected sensitive wavelengths to predict the moisture and protein contents of corn kernels, achieving high accuracy. Feature selection improved both the model accuracy and the computational complexity. Both studies showed that gradient boosting algorithms are effective and reliable tools for variable selection and prediction power in food analyses (2).

The findings of the study are supported by and consistent with those of a previous study investigating MGO-related prediction and modeling approaches. A study evaluated flavonoid–MGO reaction rates via an ANN on a limited dataset comprising some flavonoids with synthetic oversampling (39). The other ANN models were applied for predicting arginine glycation probability via a small but high-quality peptide dataset, which demonstrated that ANN-based approaches outperform simpler linear models when applied to the same data (40). There is a new study in line with our results. The DeepMGO model was developed via a comprehensive dataset including 2.262 assays and 660 compounds. They have shown that, compared with other ML/DL models, DL approaches can effectively identify novel MGO scavengers (41). In addition to these direct comparisons, broader metabolomics-based evaluations provide important context. A study reported that nonlinear models such as SVM

and ANN provided only limited improvements over traditional methods such as partial least squares (PLS), whereas random forest models showed relatively weak performance across multiple datasets (42). These findings highlight that the dataset can be more important than algorithm selection alone. Studies support the interpretation of our results, especially regarding the effectiveness of nonlinear modeling approaches for MGO-related predictions. However, they also present some limitations. Previous studies rely on relatively specialized datasets, often augmented by oversampling techniques, which may restrict generalizability. The model performance should be interpreted with consideration of dataset size, feature structure, and task types such as regression and classification. Compared with previous studies, performance metrics such as the RMSE and accuracy have been consistently reported. The results of this study, like those of other studies, clearly consider dataset characteristics. Overall, the results are consistent with the literature and offer meaningful approaches in the context of MGO estimation and modeling.

Recent studies indicate that artificial neural networks and deep learning approaches often outperform simpler models in MGO-related prediction tasks. However, these results should be interpreted with care, since small sample sizes, multicollinearity among variables, and a lack of external validation are common in chemical and food datasets. Reporting evaluation metrics, dataset characteristics, and model complexity in a transparent way is therefore important, as high predictive performance on small datasets does not always translate into generalizability. Decision trees and multilayer perceptrons remain appropriate choices for HPLC-based food analyses, but more detailed performance comparisons and clearer benchmarking against baseline models would strengthen future evaluations.

Conclusion

Machine learning algorithms were used to predict MGO formation in foods, and the model performance and variable importance levels were compared. CatBoost was the most successful model in terms of accuracy. Some vital components, such as glucose, fructose,

and saturated fat, were observed to be stronger predictors than total sugars or carbohydrates in highly processed packaged foods. In particular, fructose emerged as the most prominent variable in the Lasso, ElasticNet, and AdaBoost models, whereas glucose consistently contributed significantly across all the algorithms. Carbohydrates were identified as the dominant variables, particularly in the random forest, gradient boosting, and CatBoost models. Saturated fat significantly increased MGO formation, especially under thermal processing conditions. A comparison of the datasets revealed that carbohydrate and sugar compositions played a central role in MGO formation. Fructose and sucrose were more effective inputs for prediction than the total sugar content was. Proteins and fats provide indirect rather than direct contributions. More comprehensive and multivariate data structures yield more realistic and robust results for prediction models. Therefore, carbohydrate levels and sugar types should be considered key variables in MGO prediction. The findings of this study, together with similar applications in the literature, show that CatBoost provides good accuracy and useful variable discrimination for food safety and biochemical parameter prediction. Its success in MGO prediction suggests that this algorithm can serve as an effective tool in decision support systems on the basis of nutritional components.

XGBoost also performed well in this study, handling large datasets and reducing overfitting through built-in optimization. Together with CatBoost, it shows how machine learning can be applied to multidimensional food data and to the prediction of health-related risks.

This study has several limitations because the dataset used was derived entirely from previously published studies. In this context, although the data were homogenized, differences in laboratory protocols, sample preparation, and reporting standards in the datasets used may lead to heterogeneity according to the declared data. Although harmonization steps were applied, residual variability between sources cannot be fully eliminated. The dataset also showed considerable missingness for some nutritional variables (particularly saturated fat, fiber, and salt), which restricted the size of the multivariate subsets and likely contributed to the moderate R^2 values and the relatively

high cross-validation variance observed for the best-performing model. In addition, process-related variables such as the thermal treatment temperature, processing time, and storage conditions could not be included, as they were either not reported or reported inconsistently in the source studies. Since these parameters are known to strongly influence MGO formation, their absence represents a further source of unexplained variance. Future studies should focus on validating the pMGO model via experimentally measured MGO concentrations obtained under standardized analytical conditions, ideally from a wider range of food matrices. Expanding the predictor set to include process-related variables and exploring data augmentation strategies for underrepresented food categories may further improve the predictive accuracy and generalizability of the model.

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